

Lecture 15: Black-Scholes-Merton Model

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1 Risk-Neutral Pricing & Hedge

Using no arbitrage argument, we know that the price of a derivative at time t should be equal to the cost needed to setup a replicating portfolio for this derivative. In a binomial market, we initialized the whole pricing story with a replicating equation: find a portfolio of stocks and cash in order to match the payoff of the given derivative. By introducing an auxiliary measure, the risk-neutral measure, the price of an European derivative can be valued using backward recursion:

$$V_k = \frac{1}{1+r} \tilde{E}(V_{k+1} | \mathcal{F}_k) = \frac{1}{(1+r)^{N-k}} \tilde{E}(V_N | \mathcal{F}_k). \quad (1)$$

A binomial tree of the option prices can be finished. Once that is done, the number of shares of stocks needed at time k , Δ_k , is given by

$$\Delta_k = \frac{V_{k+1}(H) - V_{k+1}(T)}{S_{k+1}(H) - S_{k+1}(T)} = \frac{\Delta V_{k+1}}{\Delta S_{k+1}}. \quad (2)$$

In the last few lectures, we studied pricing of European options in a geometric Brownian motion market. We adopted the risk-neutral measure directly in order to get the price. We know that, in a complete market (hedging is always possible), under no arbitrage (the risk-neutral measure exists), the time t price of an European contract that pays $V(S_T)$ is given by

$$V_t = \tilde{E}(e^{-r(T-t)} V(S_T) | \mathcal{F}_t), \quad t \in [0, T]. \quad (3)$$

Furthermore, because geometric Brownian motion is a Markov process, we can actually write

$$V_t = v(t, S_t), \quad (4)$$

for some function $v(t, x)$. In the classical Black-Scholes setting, the stock price dynamics under the risk-neutral measure $\tilde{P}$ is

$$dS_t = rS_t dt + \sigma S_t dW_t, \quad S_0 > 0. \quad (5)$$

Then the function $v(t, x)$ satisfies the following partial differential equation and boundary condition:

$$\frac{\partial v}{\partial t} + rx \frac{\partial v}{\partial x} + \frac{1}{2} \sigma^2 x^2 \frac{\partial^2 v}{\partial x^2} = rv, \quad t \in [0, T] \quad (6)$$

$$v(T, x) = V(x) \quad (7)$$

However, we were never able to solve the replication question directly in the continuous setting. Martingale representation theorem only guarantees possibility of replicating, but one needs an explicit replicating strategy if he has to hedge the risk.

2 Let's Hedge

We want to derive an explicit replicating strategy for an European call option with strike $K > 0$ and maturity T . As pointed out in last lecture, the replicating/hedging part has nothing to do with probability measures. The strategy is supposed to work along any possible trajectory of stock price, no matter how you change the

probability measure. For this reason, we can just work with the actual measure here. Let us suppose the stock price process follows

$$dS_t = \alpha S_t dt + \sigma S_t dW_t, \quad S_0 > 0. \quad (8)$$

The discounted stock price process follows

$$d(e^{-rt} S_t) = e^{-rt} (-r S_t dt + dS_t) = e^{-rt} [(\alpha - r) S_t dt + \sigma S_t dW_t]. \quad (9)$$

The dynamics of any portfolio value process follows

$$dX_t = \Delta_t dS_t + r(X_t - \Delta_t S_t) dt, \quad (10)$$

whose discounted process follows

$$d(e^{-rt} X_t) = e^{-rt} (-r X_t dt + dX_t) = \Delta_t e^{-rt} [(\alpha - r) S_t dt + \sigma S_t dW_t] = \Delta_t d(e^{-rt} S_t). \quad (11)$$

This means that the discounted portfolio value changes just because the discounted stock price changes.

Let us look at what happens for the call price. We first assume that there exists a function $c(t, x)$ so that the value $c(t, x)$ is equal to the time t price of the above call if $S_t = x$. In other words, $c(t, S_t)$ is the call price at any time $t \in [0, T]$. Using Itô's formula, we have

$$\begin{aligned} dc(t, S_t) &= \frac{\partial c}{\partial t} + \frac{\partial c}{\partial x} dS_t + \frac{1}{2} \frac{\partial^2 c}{\partial x^2} (dS_t)^2 \\ &= \left(\frac{\partial c}{\partial t} + \alpha S_t \frac{\partial c}{\partial x} + \frac{1}{2} \sigma^2 S_t^2 \frac{\partial^2 c}{\partial x^2} \right) dt + \sigma S_t \frac{\partial c}{\partial x} dW_t. \end{aligned} \quad (12)$$

Let us also have a look at its discounted price process

$$\begin{aligned} d(e^{-rt} c(t, S_t)) &= e^{-rt} [-rc(t, S_t) dt + dc(t, S_t)] \\ &= e^{-rt} \left\{ \left(-rc(t, S_t) + \frac{\partial c}{\partial t} + \alpha S_t \frac{\partial c}{\partial x} + \frac{1}{2} \sigma^2 S_t^2 \frac{\partial^2 c}{\partial x^2} \right) dt + \sigma S_t \frac{\partial c}{\partial x} dW_t \right\}. \end{aligned} \quad (13)$$

In order to find a “good” process $\{\Delta_t\}_{t \in [0, T]}$ so that the resulting portfolio value X_t always matches $c(t, S_t)$, it suffices to have

$$d(e^{-rt} X_t) = d(e^{-rt} c(t, S_t)), \quad t \in [0, T] \quad (14)$$

Comparing (11) with (13), we have that

$$\Delta_t [(\alpha - r) S_t dt + \sigma S_t dW_t] = \left(-rc(t, S_t) + \frac{\partial c}{\partial t} + \alpha S_t \frac{\partial c}{\partial x} + \frac{1}{2} \sigma^2 S_t^2 \frac{\partial^2 c}{\partial x^2} \right) dt + \sigma S_t \frac{\partial c}{\partial x} dW_t. \quad (15)$$

Let us choose $\Delta_t = c_x(t, S_t)$ so that the Brownian motion W_t won't give us any trouble. Then we have an equation

$$(\alpha - r) S_t \frac{\partial c}{\partial x} = -rc(t, S_t) + \frac{\partial c}{\partial t} + \alpha S_t \frac{\partial c}{\partial x} + \frac{1}{2} \sigma^2 S_t^2 \frac{\partial^2 c}{\partial x^2}, \quad (16)$$

or in other words,

$$-rc(t, S_t) + \frac{\partial c}{\partial t} + r S_t \frac{\partial c}{\partial x} + \frac{1}{2} \sigma^2 S_t^2 \frac{\partial^2 c}{\partial x^2} = 0, \quad (17)$$

which is exactly (6). However, we did not use risk-neutral measure here.

The terminal condition for (17) is obvious:

$$c(T, x) = (x - K)^+. \quad (18)$$

We also have two bounded conditions

$$c(t, 0) = 0, \quad \lim_{x \rightarrow \infty} \frac{c(t, x)}{x} = 1. \quad (19)$$

3 Black-Scholes Formula

You have already computed call option prices with formula (3), so there is no need to solve the PDE (17). In particular, we know that

$$c(t, x) = xN(d_+) - Ke^{-r(T-t)}N(d_-), \quad t \in [0, T], x > 0 \quad (20)$$

where

$$d_{\pm} = \frac{1}{\sigma\sqrt{T-t}} \left[\log \frac{x}{K} + \left(r \pm \frac{\sigma^2}{2} \right) (T-t) \right], \quad (21)$$

and N is the cumulative standard normal distribution

$$N(y) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^y e^{-\frac{z^2}{2}} dz = \frac{1}{\sqrt{2\pi}} \int_{-y}^{\infty} e^{-\frac{z^2}{2}} dz. \quad (22)$$

We can verify the terminal conditions for $c(t, x)$. As $t \rightarrow T^-$, we have

$$\lim_{t \rightarrow T^-} d_{\pm} = \begin{cases} \infty, & x > K \\ 0, & x = K \\ -\infty, & x < K \end{cases}. \quad (23)$$

So as $t \rightarrow T^-$, $c(t, x) \rightarrow (x - K)^+$.

We can calculate the delta $c_x(t, x)$ explicitly.

$$\begin{aligned} \frac{\partial c}{\partial x} &= N(d_+) + x\phi(d_+) \frac{\partial d_+}{\partial x} - Ke^{-r(T-t)}\phi(d_-) \frac{\partial d_-}{\partial x} \\ &= N(d_+) + \frac{1}{\sigma\sqrt{T-t}} \left(\phi(d_+) - \frac{K}{x} \phi(d_-) e^{-r(T-t)} \right), \end{aligned} \quad (24)$$

where $\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$. From (21) we know that

$$d_- = d_+ - \sigma\sqrt{T-t}. \quad (25)$$

Therefore, we can show that

$$\begin{aligned} \phi(d_-) &= \phi(d_+ - \sigma\sqrt{T-t}) \\ &= \frac{1}{\sqrt{2\pi}} \exp \left\{ -\frac{(d_+ - \sigma\sqrt{T-t})^2}{2} \right\} \\ &= \frac{1}{\sqrt{2\pi}} \exp \left\{ -\frac{d_+^2 - 2d_+\sigma\sqrt{T-t} + \sigma^2(T-t)}{2} \right\} \\ &= \phi(d_+) \exp \left\{ \log \frac{x}{K} + r(T-t) \right\} \\ &= \frac{x}{K} \phi(d_+) e^{r(T-t)}. \end{aligned} \quad (26)$$

From (24) we obtain that

$$c_x(t, x) = N(d_+). \quad (27)$$